

What is Snowflake?

01
/12

Snowflake is a fully-managed cloud data platform that separates storage from compute — so you can store unlimited data cheaply and scale compute independently, paying only per second of use.

WHY IT MATTERS

- ✓ No servers to manage — Snowflake runs it all for you
- ✓ Storage & compute scale separately — no contention, no waste
- ✓ Pay-per-second billing — spin up, run, suspend, save money
- ✓ One platform for warehousing, data lakes, engineering & ML

WHERE IT RUNS

- AWS**
Amazon Web Services regions worldwide
- Azure**
Microsoft Azure cloud
- GCP**
Google Cloud Platform

“

Snowflake isn't a database you install — it's a data cloud you simply use.

— PJ'S ACADEMY

THE BIG PICTURE



AT A GLANCE

FOUNDED
2012

IPO
2020 (biggest
software IPO)

SALARY
₹18-45 LPA in India

REAL WORLD EXAMPLES

Netflix
Analyses viewing data across millions of users

Capital One
Runs risk & fraud analytics at scale of users

Instacart
Powers real-time inventory analytics

EA Games
Player behaviour & telemetry analysis

DID YOU KNOW?

The name reflects its founders' love of snow sports.

It can query petabytes without you managing a single server.

Storage and compute scale completely independently.

HOW IT WORKS



THE 3 LAYERS



EDITIONS

- › Standard
- › Enterprise
- › Business Critical
- › VPS

INTERFACES

- › Snowsight (UI)
- › SnowSQL (CLI)
- › Python / Spark
- › JDBC / ODBC

VS TRADITIONAL

- ✓ No tuning knobs
- ✓ Auto-scaling
- ✓ Pay per second

USE CASES

- 📊 Warehousing
- 🔧 Data engineering
- 🤖 ML / AI

KEY TAKEAWAYS

01

No servers to manage — Snowflake runs it all for you

02

Storage & compute scale separately — no contention, no waste

03

Pay-per-second billing — spin up, run, suspend, save money

⚡ Master Snowflake and you master the modern data stack.

QUICK REFERENCE

KEY TERMS		BILLING	
Warehouse	Compute cluster	Credit	Unit of compute
Micro-partition	Storage unit	Per-second	Billing model
EDITIONS		CLOUDS	
Standard	Entry tier	AWS	Most regions
Enterprise	90-day travel	Azure/GCP	Also supported

NEXT UP IN THE SERIES

Virtual Warehouses

The compute engines that power every query.

02

/12

Virtual Warehouses

02 /12

A virtual warehouse is a cluster of compute that runs your queries. Resize it up for heavy jobs, scale it out for many users, and auto-suspend it to stop paying when idle.

WHY IT MATTERS

- ✓ Compute is separate from storage — run 10 at once, no conflict
- ✓ Resize instantly (XS → 6XL) — no downtime, no data reload
- ✓ Auto-suspend/resume means you pay only while querying
- ✓ Multi-cluster warehouses handle concurrency spikes automatically

TWO WAYS TO SCALE

- Scale UP**
Bigger warehouse for complex queries
- Scale OUT**
More clusters for more concurrent users
- Auto-suspend**
Pause when idle to save credits



Bigger isn't always better —
right-sizing your warehouse is the real skill.

— PJ'S ACADEMY

THE BIG PICTURE



Each size up doubles both compute power and credits/hour.

AT A GLANCE

SIZES
XS to 6XL

SCALING
Up or Out

SUSPEND
60s recommended

REAL WORLD EXAMPLES

Retail
Scale out for Black Friday query spikes

BI teams
Separate warehouse per department

Data science
Large warehouse for heavy ML feature jobs

ETL
Dedicated warehouse for nightly loads

DID YOU KNOW?

A suspended warehouse costs \$0 — only storage is billed.

Each size step doubles credits: XS=1, S=2, M=4, L=8/hr.

Multi-cluster is an Enterprise-edition feature.

USING WAREHOUSES



SIZE — CREDITS/HR



SETTINGS

- › WAREHOUSE_SIZE
- › AUTO_SUSPEND
- › AUTO_RESUME
- › MIN/MAX_CLUSTER

BEST PRACTICE

- › Suspend at 60s
- › Start small
- › Separate by workload

SCALE UP

- Complex query
- Big joins
- Resize bigger

SCALE OUT

- Many users
- Concurrency
- Add clusters

KEY TAKEAWAYS

- 01**
Compute is separate from storage — run 10 at once, no conflict
- 02**
Resize instantly (XS → 6XL) — no downtime, no data reload
- 03**
Auto-suspend/resume means you pay only while querying

⚡ Control your warehouses well and you control your bill.

QUICK REFERENCE

SIZES		COMMANDS	
XSMALL	1 credit/hr	CREATE WAREHOUSE	New WH
6XLARGE	512 credits/hr	ALTER ... SET	Resize
SCALING		SAVING	
Scale up	Resize	AUTO_SUSPEND	Pause idle
Scale out	Multi-cluster	AUTO_RESUME	Wake on query

NEXT UP IN THE SERIES

Micro-Partitions

How Snowflake stores and prunes data for speed.

03

/12

Micro-Partitions

03 /12

Snowflake automatically stores data in compressed, columnar micro-partitions (50–500 MB each). Metadata on every partition lets Snowflake skip irrelevant data — this is 'pruning', and it's why queries are fast.

WHY IT MATTERS

- ✓ Automatic — you never create or manage partitions
- ✓ Columnar + compressed = less storage, faster scans
- ✓ Metadata enables pruning — skip data that can't match
- ✓ Clustering keeps related data together for big tables

KEY IDEAS

- Micro-partition**
50–500 MB compressed columnar unit
- Pruning**
Skip partitions that can't match a filter
- Clustering**
How well-sorted data is across partitions

“

Snowflake reads less data, not faster data — that's the secret to its speed.

— PJ'S ACADEMY

THE BIG PICTURE



Min/max metadata per partition powers automatic pruning.

AT A GLANCE

SIZE
50–500 MB

FORMAT
Columnar

MANAGED
Automatic

REAL WORLD EXAMPLES

Analytics
Filter billions of rows, scan only a few partitions

Time-series
Date filters prune to relevant days instantly

Large tables
Clustering keys keep TB-scale tables fast

Cost
Less scanned = fewer credits burned

DID YOU KNOW?

- Good pruning can skip 99% of a table for a selective query.
- Clustering keys only help on very large (>1TB) tables.
- Each partition stores min, max, and count per column.

THE FLOW



PRUNING IN ACTION



AUTOMATIC

- › Partitioning
- › Compression
- › Metadata
- › Pruning

YOU CONTROL

- › Clustering keys
- › Reclustering
- › Table design

GOOD FOR

- ✓ Selective filters
- ✓ Date ranges
- ✓ Big tables

WATCH

- ! Over-clustering
- ! Tiny tables
- ! Cost of recluster

KEY TAKEAWAYS

01

Automatic — you never create or manage partitions

02

Columnar + compressed = less storage, faster scans

03

Metadata enables pruning — skip data that can't match

⚡ Understanding pruning is how you write fast, cheap queries.

QUICK REFERENCE

STORAGE		METADATA	
Columnar	Per-column	Min/Max	Per column
Compressed	Auto	Count	Per partition
SPEED		SCALE	
Pruning	Skip data	Micro-part	50–500MB
Clustering	Sort order	>1TB	Add clustering

NEXT UP IN THE SERIES

Loading Data

Get data into Snowflake with COPY and Snowpipe.

04

/12

Loading Data

04
/12

Get data in with stages + COPY INTO for bulk loads, or Snowpipe for continuous auto-loading. Snowflake handles CSV, JSON, Parquet, Avro and more — and parallelises loading automatically.

WHY IT MATTERS

- ✓ COPY INTO bulk-loads files fast, in parallel
- ✓ Snowpipe auto-loads the moment files arrive — serverless
- ✓ Stages abstract where files live (S3, Azure, GCS, internal)
- ✓ Load structured AND semi-structured data the same way

TWO LOAD MODES

- COPY INTO**
Bulk load — uses a warehouse
- Snowpipe**
Continuous auto-load — serverless
- Stages**
Pointers to where your files live



Batch or streaming, structured or JSON —
Snowflake loads it all the same way.

— PJ'S ACADEMY

THE BIG PICTURE



Snowpipe uses cloud notifications to load files automatically.

AT A GLANCE

- FORMATS**
CSV, JSON, Parquet
- CONTINUOUS**
Snowpipe
- PARALLEL**
Automatic

REAL WORLD EXAMPLES

- Apps**
Snowpipe streams event logs in near real-time
- IoT**
Sensor files auto-load as they land
- BI**
Nightly COPY loads for dashboards
- Kafka**
Snowflake Kafka connector streams topics

DID YOU KNOW?

- Snowpipe is serverless — no warehouse needed to load.
- COPY INTO can skip bad rows with `ON_ERROR=CONTINUE`.
- You can `VALIDATE` a load before committing it.

LOADING STEPS



STAGE TYPES



FORMATS

- CSV
- JSON
- Parquet
- Avro / ORC

ON_ERROR

- ABORT (default)
- CONTINUE
- SKIP_FILE

BULK (COPY)

- Big batches
- Uses warehouse
- Manual/scheduled

STREAM (PIPE)

- Continuous
- Serverless
- Auto-ingest

KEY TAKEAWAYS

01

COPY INTO bulk-loads files fast, in parallel

02

Snowpipe auto-loads the moment files arrive — serverless

03

Stages abstract where files live (S3, Azure, GCS, internal)

⚡ Loading well means clean, reliable data for everything downstream.

QUICK REFERENCE

LOAD		STAGES	
COPY INTO	Bulk	Internal	@stage
Snowpipe	Stream	External	S3/Azure
FORMATS		ERRORS	
CSV	Structured	CONTINUE	Skip bad rows
JSON	VARIANT	VALIDATION	Dry run

NEXT UP IN THE SERIES

Semi-Structured Data

Query JSON as easily as tables with VARIANT.

05

/12



SNOWFLAKE MASTERY AWAITS...

12 visual concepts · 20 projects · full SnowPro Core cert prep

DM
"SNOWFLAKE"

SAVE IT
Save for later

SHARE IT
Help others learn

FOLLOW
@pjsacademy.datascience

LEARN. BUILD. CERTIFY.
Snowflake, mastered.

📚 BONUS RESOURCE ✓ 60+ SnowPro Practice Qs ✓ 20 Hands-On Projects ✓ SnowPro Core Cheatsheet

pjsacademy.com

Semi-Structured

05 /12

Snowflake's VARIANT type stores JSON, Avro, and XML natively — no flattening needed up front. Query nested fields with path notation and explode arrays into rows with FLATTEN.

WHY IT MATTERS

- ✓ Store raw JSON without designing a schema first
- ✓ Query nested fields as fast as structured columns
- ✓ FLATTEN turns arrays into queryable rows
- ✓ Schema-on-read — adapt as your data evolves

KEY TOOLS

VARIANT
Column type that holds any JSON

Path notation
col.field.subfield.type

FLATTEN
Explode arrays into rows

“JSON isn't second-class in Snowflake — it queries just like a table.”

— PJ'S ACADEMY

THE BIG PICTURE

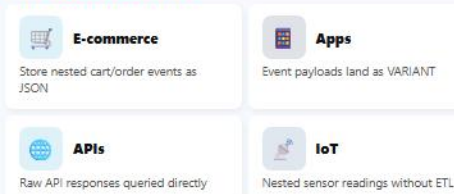


Access nested values with colon paths and :: casting.

AT A GLANCE



REAL WORLD EXAMPLES



DID YOU KNOW?

- A VARIANT value can hold up to 16 MB compressed.
- Snowflake auto-optimises frequently-queried JSON paths.
- FLATTEN is a table function used with LATERAL.

QUERYING JSON



PATH SYNTAX



FUNCTIONS

- » FLATTEN
- » PARSE_JSON
- » OBJECT_KEYS
- » ARRAY_SIZE

CASTS

- » ::STRING
- » ::NUMBER
- » ::BOOLEAN
- » ::TIMESTAMP

STORE

- ✓ No schema first
- ✓ Raw JSON
- ✓ Evolves freely

QUERY

- Path notation
- FLATTEN arrays
- Cast types

KEY TAKEAWAYS

01

Store raw JSON without designing a schema first

02

Query nested fields as fast as structured columns

03

FLATTEN turns arrays into queryable rows

⚡ Semi-structured mastery unlocks modern app & event data.

QUICK REFERENCE

TYPE		ACCESS	
VARIANT	Any JSON	col:path	Navigate
OBJECT	Key-value	::type	Cast
EXPLODE		PARSE	
FLATTEN	Arrays→rows	PARSE_JSON	String→JSON
LATERAL	Join	TO_JSON	Back

NEXT UP IN THE SERIES

SQL Transformations

Window functions, QUALIFY, and Snowflake SQL gems.

06

/12

SQL Power

06
/12

Snowflake SQL includes powerful extras: window functions with QUALIFY, PIVOT/UNPIVOT, approximate functions for speed, and LATERAL joins — everything you need to transform data at scale.

WHY IT MATTERS

- ✓ QUALIFY filters window results without a subquery
- ✓ Approximate functions give instant answers on huge data
- ✓ PIVOT/UNPIVOT reshape data in one statement
- ✓ It's ANSI SQL plus — familiar but more powerful

SIGNATURE FEATURES

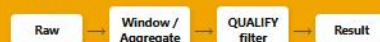
- QUALIFY**
Filter window functions inline
- PIVOT**
Rows to columns and back
- APPROX_***
Fast estimates on big data



QUALIFY is the one-line trick that replaces messy nested subqueries.

— PJ'S ACADEMY

THE BIG PICTURE



QUALIFY is to window functions what HAVING is to GROUP BY.

AT A GLANCE

- WINDOWS + QUALIFY
- SPEED APPROX funcs
- RESHAPE PIVOT

REAL WORLD EXAMPLES

- Ranking**
Top-N per group in one clean query
- Analytics**
Running totals & moving averages
- Big data**
APPROX_COUNT_DISTINCT on billions of rows
- Reports**
PIVOT monthly data into columns

DID YOU KNOW?

- APPROX_COUNT_DISTINCT uses HyperLogLog — fast & tiny.
- QUALIFY is a Snowflake extension not in standard SQL.
- SAMPLE lets you query a random % of a table instantly.

TRANSFORM FLOW



TOP-3 PER GROUP



WINDOW FN

- › ROW_NUMBER
- › RANK
- › LAG / LEAD
- › SUM OVER

RESHAPE

- › PIVOT
- › UNPIVOT
- › LATERAL
- › FLATTEN

APPROX

- COUNT_DISTINCT
- PERCENTILE
- TOP_K

CONDITIONAL

- IFF
- DECODE
- CASE

KEY TAKEAWAYS

01

QUALIFY filters window results without a subquery

02

Approximate functions give instant answers on huge data

03

PIVOT/UNPIVOT reshape data in one statement

Great SQL is the core skill every data engineer is hired for.

QUICK REFERENCE

WINDOWS		RESHAPE	
OVER()	Partition	PIVOT	Rows→cols
QUALIFY	Filter	UNPIVOT	Cols→rows
FAST		LOGIC	
APPROX_*	Estimates	IFF	Ternary
SAMPLE	Random %	CASE	Multi-branch

NEXT UP IN THE SERIES

Time Travel

Query and restore data from the past.

07

/12

Time Travel

07 /12

Time Travel lets you query data as it existed in the past — up to 90 days on Enterprise. Recover from bad updates, deletes, or drops instantly with AT/BEFORE and UNDROP. Fail-safe adds 7 more days of Snowflake-managed recovery.

WHY IT MATTERS

- ✓ Query data as it was seconds, hours, or days ago
- ✓ UNDROP a table, schema, or database you deleted by mistake
- ✓ Recover from a bad UPDATE without a backup
- ✓ Fail-safe adds 7 days of last-resort recovery

RECOVERY TOOLS

- Time Travel**
Query/restore up to 90 days back
- UNDROP**
Instantly restore dropped objects
- Fail-safe**
7-day Snowflake-managed safety net

“

The best backup is the one you never had to set up — that's Time Travel.

— PJ'S ACADEMY

THE BIG PICTURE



Fail-safe is support-only and starts after Time Travel expires.

AT A GLANCE

- MAX 90 days**
- DEFAULT 1 day**
- FAIL-SAFE 7 days**

REAL WORLD EXAMPLES

- Oops**
Recover a table dropped by accident
- Audit**
See exactly what data looked like last week
- Testing**
Clone a point-in-time state for QA
- DR**
Restore after a bad ETL run

DID YOU KNOW?

- Fail-safe can't be queried — only Snowflake support recovers it.
- 90-day Time Travel needs Enterprise edition; default is 1 day.
- UNDROP works on tables, schemas, AND whole databases.

RECOVERY FLOW



RECOVERY WAYS



SYNTAX

- AT(OFFSET=>-n)
- AT(TIMESTAMP=>..)
- BEFORE(STATEMENT=>..)
- UNDROP

RETENTION

- Standard: 1 day
- Enterprise: 90
- Transient: 0-1

TIME TRAVEL

- ✓ You query it
- ✓ Up to 90 days
- ✓ Configurable

FAIL-SAFE

- ! Support only
- ! 7 days fixed
- ! Not queryable

KEY TAKEAWAYS

01

Query data as it was seconds, hours, or days ago

02

UNDROP a table, schema, or database you deleted by mistake

03

Recover from a bad UPDATE without a backup

⚡ Time Travel turns 'we lost data' into 'one query fixes it'.

QUICK REFERENCE

QUERY PAST		RESTORE	
AT(OFFSET)	Seconds ago	UNDROP	Bring back
BEFORE(STMT)	Pre-query	CLONE AT	Point-in-time
LIMITS		SAFETY	
Default	1 day	Fail-safe	7 days
Max	90 days	Support	Only

NEXT UP IN THE SERIES

Zero-Copy Cloning

Instant dev/test copies with no storage cost.

08

/12

Zero-Copy Cloning

08 /12

CREATE ... CLONE makes an instant, writable copy of a table, schema, or entire database — with no storage cost until the clone changes. Perfect for dev, test, and experimentation environments.

WHY IT MATTERS

- ✓ Clone a whole database in seconds, not hours
- ✓ Zero storage cost until the clone diverges
- ✓ Give every developer a full-data sandbox instantly
- ✓ Combine with Time Travel to clone a past state

WHAT YOU CAN CLONE

- Table**
A single table's data & structure
- Schema**
All objects in a schema
- Database**
An entire database instantly

“

Why copy terabytes when you can clone them in seconds for free?

— PJ'S ACADEMY

THE BIG PICTURE



Only changed data consumes storage — the rest is shared.

AT A GLANCE

SPEED
Seconds

COST
₹0 until changed

SCOPE
Table—Database

REAL WORLD EXAMPLES

Dev
Per-developer sandbox with full prod data

QA
Fresh test env for every release

Analytics
Experiment without touching production

Training
Give learners real data safely

DID YOU KNOW?

A clone costs nothing until you modify its data.

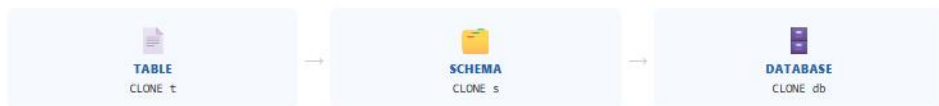
You can clone a table AS IT WAS via Time Travel.

Clones are fully independent — write freely.

CLONING FLOW



CLONE TARGETS



SYNTAX

- » CREATE TABLE ... CLONE
- » CREATE SCHEMA ... CLONE
- » CREATE DATABASE ... CLONE
- » + AT/BEFORE

USE CASES

- » Dev sandboxes
- » QA environments
- » Experiments

BENEFITS

- ✓ Instant
- ✓ No storage cost
- ✓ Independent

PAIR WITH

- Time Travel
- CI/CD
- Data sharing

KEY TAKEAWAYS

01

Clone a whole database in seconds, not hours

02

Zero storage cost until the clone diverges

03

Give every developer a full-data sandbox instantly

⚡ Zero-copy cloning is the feature that makes every team faster.

QUICK REFERENCE

CLONE		COST	
TABLE	One table	Create	Free
DATABASE	Everything	Diverge	Pay changes
TIME		USE	
CLONE AT	Past state	Dev	Sandbox
BEFORE	Pre-query	QA	Test env

NEXT UP IN THE SERIES

Snowpark

Run Python inside Snowflake — no data movement.

09

/12

Snowpark & Python

09 /12

Snowpark lets you write Python, Java, or Scala that runs inside Snowflake. Use a pandas-like DataFrame API, deploy UDFs and stored procedures, and even train ML models where your data lives — zero data movement.

WHY IT MATTERS

- ✓ Write Python that runs on Snowflake compute
- ✓ DataFrame API feels like pandas but scales massively
- ✓ Deploy UDFs so ML predictions run inside SQL
- ✓ No data egress — process where the data lives

WHAT YOU BUILD

- DataFrames**
pandas-like API on Snowflake compute
- UDFs**
Python functions callable from SQL
- Procedures**
Full procedural logic in Python

“Bring your code to the data — not the data to your code.”

— PJ'S ACADEMY

THE BIG PICTURE



Snowpark pushes computation down to Snowflake's engine.

AT A GLANCE

LANGS Python, Java, Scala	API DataFrame	EGRESS None
-------------------------------------	-------------------------	-----------------------

REAL WORLD EXAMPLES

ML Train & serve models inside Snowflake	ETL Complex transforms in Python
Features Feature engineering at scale	UDFs Custom logic callable from any query

DID YOU KNOW?

- Data never leaves Snowflake — great for security & compliance.
- The Snowpark DataFrame API mirrors pandas closely.
- A Python UDF turns ML inference into a SQL function.

SNOWPARK FLOW



DATAFRAME OPS



LANGUAGES

- › Python
- › Java
- › Scala
- › SQL Interop

DEPLOY

- › UDFs
- › UDTFs
- › Stored procs
- › ML models

DATAFRAME

- filter
- group_by
- join

BENEFITS

- ✓ No egress
- ✓ Scales
- ✓ Familiar

KEY TAKEAWAYS

01

Write Python that runs on Snowflake compute

02

DataFrame API feels like pandas but scales massively

03

Deploy UDFs so ML predictions run inside SQL

⚡ Snowpark is where data engineering meets data science.

QUICK REFERENCE

API		DEPLOY	
Session	Connect	UDF	SQL callable
DataFrame	Transform	Procedure	Logic
LANGS		WIN	
Python	Most common	No egress	Secure
Java/Scala	Also	Push-down	Fast

NEXT UP IN THE SERIES

Streams & Tasks

Build real-time pipelines inside Snowflake.

10

/12

Streams & Tasks

10 / 12

Streams track changes (inserts, updates, deletes) on a table — Snowflake-native CDC. Tasks run SQL on a schedule or in a chain. Together they build real-time ELT pipelines with no external orchestrator.

WHY IT MATTERS

- ✓ Streams capture row-level changes automatically (CDC)
- ✓ Tasks schedule or chain SQL — a pipeline DAG
- ✓ Combine for near-real-time ELT, no external tools
- ✓ Trigger tasks only when a stream has new data

THE BUILDING BLOCKS

- Stream**
Tracks changes on a table (CDC)
- Task**
Scheduled or chained SQL execution
- Task DAG**
Chain tasks into a pipeline

“

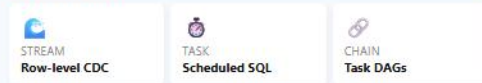
Streams + Tasks = a real-time data pipeline without leaving Snowflake.

— PJ'S ACADEMY

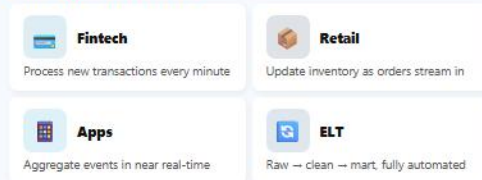
THE BIG PICTURE



AT A GLANCE



REAL WORLD EXAMPLES



DID YOU KNOW?

- A stream stores only the **CHANGES**, not a full copy.
- Tasks can run as often as every minute — or serverless.
- Chained tasks form a DAG — a whole pipeline in Snowflake.

PIPELINE FLOW



PIPELINE STAGES



STREAMS

- › Standard
- › Append-only
- › Insert-only
- › `SYSTEM$STREAM_HAS_DATA`

TASKS

- › SCHEDULE
- › AFTER (chain)
- › WHEN condition
- › Serverless

STREAM

- Tracks CDC
- Insert/upd/del
- Consumed once

TASK

- Runs SQL
- Scheduled
- Chainable

KEY TAKEAWAYS

01

Streams capture row-level changes automatically (CDC)

02

Tasks schedule or chain SQL — a pipeline DAG

03

Combine for near-real-time ELT, no external tools

⚡ This is how modern real-time data platforms are built.

QUICK REFERENCE

STREAM		TASK	
CREATE STREAM	On table	SCHEDULE	Cron/interval
CDC	Track changes	AFTER	Chain
TRIGGER		PATTERN	
STREAM_HAS_DATA	Run if new	Raw→Clean	Task 1
WHEN	Condition	Clean→Mart	Task 2

NEXT UP IN THE SERIES

Security & RBAC

Roles, masking, and enterprise governance.

11

/ 12

Security & RBAC

11 /12

Snowflake secures data with Role-Based Access Control, dynamic data masking, row-level security, and always-on encryption. Grant privileges to roles, roles to users — least-privilege by design.

WHY IT MATTERS

- ✓ RBAC — grant privileges to roles, roles to users
- ✓ Dynamic masking hides PII from unauthorised roles
- ✓ Row-level security filters rows per user
- ✓ Encryption at rest & in transit is always on

SECURITY LAYERS

RBAC
Role-based access control hierarchy

Masking
Hide sensitive columns dynamically

Encryption
AES-256, automatic, always on

“Least privilege isn't paranoia — it's professional data governance.”
— PJ'S ACADEMY

THE BIG PICTURE

Privilege → Role → User → Access

ACCOUNTADMIN sits on top; build custom roles under SYSADMIN.

AT A GLANCE

MODEL
RBAC + DAC

ENCRYPT
AES-256

MASK
Dynamic

REAL WORLD EXAMPLES

Banking
Mask account numbers from analysts

Healthcare
Row-level access to patient data

Global
Region-restricted data via policies

IN India
DPDP-compliant PAN/Aadhaar masking

DID YOU KNOW?

Encryption is automatic — keys rotate every 30 days.

One masking policy can protect a column across all queries.

Best practice: never use ACCOUNTADMIN for daily work.

GOVERNANCE FLOW



ROLE HIERARCHY



SYSTEM ROLES

- » ACCOUNTADMIN
- » SECURITYADMIN
- » SYSADMIN
- » PUBLIC

PROTECTION

- » Masking policy
- » Row access policy
- » Object tags

RBAC

- Grant to roles
- Roles to users
- Inheritance

SECURE

- ✓ Masking
- ✓ Row-level
- ✓ Encryption

KEY TAKEAWAYS

01
RBAC — grant privileges to roles, roles to users

02
Dynamic masking hides PII from unauthorised roles

03
Row-level security filters rows per user

⚡ Governance skills are what enterprises pay senior engineers for.

QUICK REFERENCE

ROLES		MASK	
System	Built-in	Policy	Per column
Custom	Under SYSADMIN	Dynamic	By role
ROW		CRYPTO	
Row policy	Filter rows	At rest	AES-256
Per user	Context	In transit	TLS

NEXT UP IN THE SERIES

Cost & Certification

Optimise spend and pass SnowPro Core.

12 /12

Cost & SnowPro

12 /12

Run Snowflake efficiently with resource monitors, right-sized warehouses, and query tuning — then prove your skills with the SnowPro Core certification. This is the skill combo that lands ₹18–45 LPA data roles.

WHY IT MATTERS

- ✓ Resource monitors cap spend before it surprises you
- ✓ Right-sizing warehouses is the #1 cost lever
- ✓ ACCOUNT_USAGE reveals your most expensive queries
- ✓ SnowPro Core certification proves it all to employers

THE FINAL SKILLS

- Cost control**
Monitors, sizing, query tuning
- Monitoring**
ACCOUNT_USAGE views
- SnowPro**
Industry-recognised certification

“

Running Snowflake cheaply is the skill that gets data engineers promoted.

— PJ'S ACADEMY

THE BIG PICTURE

Monitor → Optimise → Certify → Get hired

SnowPro Core validates all 6 exam domains you've now learned.

AT A GLANCE

- CONTROL**
Resource monitors
- CERT**
SnowPro Core
- SALARY**
₹18–45 LPA

REAL WORLD EXAMPLES

- FinOps**
Cap monthly credits with monitors
- Tuning**
Find & fix expensive queries
- Careers**
SnowPro-certified engineers in demand
- Growth**
Cert + projects = job offers

DID YOU KNOW?

- A resource monitor can auto-suspend at a credit limit.
- SnowPro Core is valid for 2 years and costs \$175.
- ACCOUNT_USAGE keeps up to 365 days of history.

YOUR PATH



EXAM DOMAINS



COST TOOLS

- › Resource monitors
- › Auto-suspend
- › Right-sizing
- › Query profile

MONITORING

- › ACCOUNT_USAGE
- › QUERY_HISTORY
- › METERING_HISTORY

SAVE

- Suspend idle
- Right-size
- Cap credits

CERTIFY

- 60+ practice Qs
- Mock exams
- Pass!

KEY TAKEAWAYS

01

Resource monitors cap spend before it surprises you

02

Right-sizing warehouses is the #1 cost lever

03

ACCOUNT_USAGE reveals your most expensive queries

⚡ You now know Snowflake end to end — go get certified & hired!

QUICK REFERENCE

COST		WATCH	
Monitor	Cap spend	QUERY_HISTORY	Slow queries
Right-size	Save credits	METERING	Credits
EXAM		NEXT	
SnowPro Core	\$175	Advanced	Data Engineer
Valid	2 years	Career	₹18–45 LPA

NEXT UP IN THE SERIES

You Did It!

Enrol at pjsacademy.com to go from here to certified.

01

/12



SNOWFLAKE MASTERY AWAITS...

12 visual concepts • 20 projects • full SnowPro Core cert prep

DM
"SNOWFLAKE"

SAVE IT
Save for later

SHARE IT
Help others learn

FOLLOW
[@pjsacademy.datascience](https://pjsacademy.datascience)

LEARN. BUILD. CERTIFY.
Snowflake, mastered.

📚 **BONUS RESOURCE** ✓ 60+ SnowPro Practice Qs ✓ 20 Hands-On Projects ✓ SnowPro Core Cheatsheet

pjsacademy.com